

Pre-Test Draft

Target Information and Profile System (TIPS) User Requirements

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Prepared by

**U S WEST Advanced Technologies
Carnegie Group Inc.**

Abstract

This document provides accurate, detailed descriptions of user problems, objectives, and requirements for the Target Information and Profile system.

It defines all hardware and software requirements, but does not describe any design, verification, or project management details, except for required design constraints. It limits the range of valid solutions, but does not project any particular design.

Version Notice

All revisions made to this document are listed below in chronological order. The release number of this document does not relate to the release number of the application:

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To Obtain Copies, Contact

Document Coordinator
U S WEST Communications, Inc.
1475 Lawrence Street, 5th Floor
Denver, Colorado 80202
(303) 595-2826

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1. Executive Summary

This section briefly describes the Target Information and Profile System.

1.1 Problem Summary

Target Information and Profile System (TIPS) will provide sales assistance and tips to the Small Business Market Unit (SBMU) Customer Contact Employees (CCEs). The overall objective of TIPS is to organize, represent and present sales, product and marketing knowledge in a form that can be shared, collected, communicated and utilized effectively by CCEs and other departments involved. TIPS is designed to help the CCEs sell broader and deeper. That is, help them to sell across the entire range of products that USWEST offers small businesses, and enable them to sell more products in a single contact. The way it will be done is to organize and present the marketing knowledge.

In the initial design and implementation, TIPS is envisioned to be an intelligent sales tool integrated into Sales and Service Negotiation and Quality (SSNQ) system. It will generate customized product recommendations based on the telecommunication needs of the customers.

1.2 User Requirements

TIPS will collect and display profile information of customers. Based on the customer information, it will generate customized product recommendations. Other sales support information such as benefits, counters and how the recommended products work.

TIPS will display advertising campaigns and allow CCE to record responses from the customers. The data collected may be used as a reference for the effectiveness of advertising campaigns. Other functions of TIPS include displaying special product promotions and office blitzes.

1.3 Recommendations

This project will develop an intelligent sales tool. Its immediate results are the collection of salient customer profile and market conditions, data which is not currently available. The quality of the knowledge base will improve continuously as more knowledge is available and is fed back to the knowledge base. It will become a useful source of conducting further market research and analysis. The knowledge gathered will serve to enhance the effectiveness of CCEs in consultative selling.

2. Introduction

This section briefly describes TIPS.

2.1 Purpose

This document identifies the needs of Customer Contact Employees (CCEs) for information to conduct consultative sales and of Marketing Units to conduct market analysis. It also identifies those requirements that the TIPS must meet and must incorporate to effectively address those problems.

2.2 Authors

The authors of this document are Kathy Eckman, Senior Engineer of Carnegie Group Inc. and Yat-Sang Hung, Member of Technical Staff, Standard Services and Human Factors Development, US WEST Advanced Technologies. Documentation support was provided by James Beetem, Information Development, Information Technologies (IT).

2.3 Project Description

The objective of the TIPS is organization, representation and presentation of sales, product and marketing knowledge in a form that can be shared, collected, communicated and utilized effectively by CCEs and other departments involved.

In the initial design and implementation, TIPS is envisioned as an intelligent sales tool integrated with the Sales and Service Negotiation and Quality (SSNQ) System. Initially, Small Business Service (SBS) CCEs will use it to sell broader and deeper by providing quality consultative services to customers. TIPS will generate customized product recommendations and sales tips based on the telecommunication needs of the customers.

2.4 Project Scope

The project designers and developers will perform the following activities within the scope of the project:

- Conduct knowledge acquisition sessions with the subject matter experts to elicit the knowledge necessary to build the knowledge base for the underlying engine; it is assumed that the subject matter experts will be able to supply the necessary knowledge in a timely and consistent fashion.

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- Organize the knowledge elicited from the subject matter experts and other sources — product statistics — and implement a system capable of generating recommendations from such knowledge using an appropriate technology.
- Provide a simple interface where knowledge can be updated easily by the client organization.
- Incorporate specified components of a sales tips helper to add value to the system.
- Design a user interface prototype for all the components of TIPS, implementation of the user interface will be provided through SSNQ.
- Specify the type of data services needed, implementation of data services will be provided through SSNQ.

The project will not perform the following activities not within the scope of the project:

- Conduct marketing research for SBS to determine the relationship between product recommendation and customer information and needs as defined above.
- Conduct marketing or any related research to build or enhance knowledge for the knowledge base.
- Maintain any changes in product information, advertisements, office blitz, blue-light specials and any other data that need update.
- Interface with legacy systems.
- Conduct any research on the data that TIPS gathers.

2.5 Assumptions and Dependencies

It is assumed that SSNQ can provide the necessary resources for data services and user interface component development in a timely and consistent manner between June and November of 1993. Absence of such support will impact the quality and schedule of the project. The resources needed will be estimated after the engineering team of Carnegie Group Inc. finishes their first review of the user requirements.

3. Current Operations

3.1 Description of Current System

For details on the current system operations, refer to the *Sales and Service Negotiation and Quality (SSNQ) User Requirements*.

3.2 Statement of Problem or Need

For a description of the TIPS problem statement and needs, refer to the *Sales and Service Negotiation and Quality (SSNQ) User Requirements*.

3.3 Client Evaluation of Current System

For details on the current system description, refer to the *Sales and Service Negotiation and Quality (SSNQ) User Requirements*.

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4. Definition of Requirements

4.1 Functional Requirements

The client initially requires that TIPS incorporate the following functions in the initial phase:

The following functions will be completed by Carnegie Group Inc. with the assistance of the Knowledge-Based Systems Group of US WEST, as indicated in their statement of work:

- Collect pertinent customer/account profile and market information necessary for generating recommendations; the system should gather, display and store profile information that can be utilized by the CCEs and marketing organization. The amount of profile information collected should be kept to a level so the CCEs have sufficient time to complete in a customer contact.
- Identify with reasonable accuracy the products that the customers would typically buy, based on customer/account profile/needs and market information. The system will display recommendations for products with nonrecurring and monthly rates (a la Carte) for each product recommended on a line-by-line basis. The recommendations generated should take into account the existing services of the customers.
- Provide benefits, counters, transition and closing cues and applications for product recommendations based on customer profile/needs and market information. This includes the capability for providing descriptions of benefits for each product recommended tailored specifically to the customer and to the different situations when a customer can use a recommended product. This also includes information on how each product works.
- Allow modifications to knowledge base contents with the long-term direction that it can be performed by marketing and product management. TIPS has to provide the file structure specification of the profile contents and sales information so that marketing or product management can conduct analysis on the data.

The following functions will be completed by Information Technologies of US WEST in the initial or subsequent phase:

- Provide notification of applicable specials and promotions. Information of specials and promotions will be maintained by personnel from SBS. Office Blitz will also be included.
- Provide help facility for the functions of TIPS. The help facility will be designed in a manner consistent with SSNQ help function.

The following functions will be completed by Information Technologies of US WEST as SSNQ Functional Requirements in the initial or subsequent phase:

- Provide information on advertising campaigns and collect customer feedback on their effectiveness. Advertising information will be kept and updated by personnel from SBS. The feedback collected will be a simple count of the number of customers who have seen specific advertisements when they are asked by the CCEs.
- Provide local calling area information.
- Display new product information.
- Provide conversion dates for central offices (product availability and switch upgrades).
- Show frequency of products bought across customer categories.

The following functions are currently under review by SBS and IT:

- Provide access from the TIPS knowledge base to external systems, for example to the DMP or to general information stored in various outside data stores. (Enhancement of DMP has to be done)
- Provide a weekly summary of bulletins. (Maintenance Issue)
- Provide information on past US WEST toll for 12 months. (System Issue)
- Display future product availability.
- Provide current affairs in local region. (Maintenance Issue)
- Provide an intra-departmental directory. (Maintenance Issue)
- Provide a catalog of products showing USOCs and their English meanings. (DMP Function)
- Provide daily feedback and sales profile information for CCEs. (Assistant Coach Function)
- Provide a feedback channel from the CCE to marketing and product management. (COMPASS Platform)
- Provide a brief description of customer business procedures.

Knowledge validation sessions with subject matter experts (CCEs, marketing and product management) will be conducted at appropriate times to review and enhance the knowledge that was acquired at various stages in the process. A series of sales contact scenarios will be constructed by the subject matter experts and will be used as the testing material for the engine and knowledge base developed. These quality control processes will be implemented to improve the quality and validity of the knowledge base and the engine developed for TIPS.

4.2 User Interface Requirements

- Allow the CCE to invoke TIPS through SSNQ.
- Populate fields automatically from the data associated with local data stores and data services provided by SSNQ.

- Display customized product recommendations, taking into account the existing products utilized by the customers.
- Display price information (recurring and non-recurring) on the products recommended by the system.
- Provide the CCEs with a seamless presentation layer where all system interactions remain invisible.
- Allow the CCEs to interact with the system by pointing and clicking with a mouse, or typing at their keyboard if they prefer. The keyboard interaction will involve only screen commands. Actual data entry must rely on point and click as much as possible to reduce errors and preserve the quality of the service request.
- Collect information on customer profile and needs required for the knowledge base at a level that will not hinder the consultative activities of CCEs.

4.3 Performance Requirements

As a subcomponent of SSNQ, TIPS must meet the basic performance requirements for that system, as documented in the Sales and Service Negotiation and Quality (SSNQ) User Requirements. In particular, TIPS must meet the following basic requirements:

- TIPS must support terminal requests from more than 600 SBMU CCEs simultaneously.
- TIPS must be available on-line between 7:00 A.M. through 7:00 P.M. Monday through Friday, and between 7:00 A.M. through 5:00 P.M. on Saturday. TIPS must be 100% operational during these times. Supporting systems must likewise be 100% available.
- TIPS will have a 100% operational availability for the interface. Most of the data required by TIPS is retrieved through SSNQ, which has an identical operational availability requirement.

TIPS will be in use for only a small portion of a typical SSNQ customer call; its performance impact on SSNQ and on the overall environment will be minimal.

TIPS uses Local Database to store customer information. This database is accessed at most twice per call (once for retrieval, once for storage).

4.4 System Requirements

TIPS is launched through SSNQ, so security is provided through SSNQ.

All access from TIPS to other US WEST data systems will be channeled through SSNQ. SSNQ will provide transparent and secure access to the needed information.

TIPS planning must address the following areas of backup and recovery:

- normal system backup and recovery handled by Distributed Services Operations Management (DSOM)
- database backup and recovery handled by Data Recovery Management (DRM)

TIPS will perform system backup and recovery consistent with SSNQ. SSNQ is consistent with the backup and recovery approach for all applications launched from COMPASS.

TIPS will determine the requirements for its local database backup and recovery as the details of its database requirements become clearer. Database backup and recovery mechanisms will be consistent with those adopted for SSNQ. TIPS must design both backup and recovery to meet system availability requirements specified in the Performance Requirements section of this document.

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4.5 Equipment Requirements

Equipment Requirements are the same as SSNQ. For details, refer to the *Sales and Service Negotiation and Quality (SSNQ) User Requirements*.

4.6 Printing Requirements

Printing Requirements will be the same as SSNQ. For details, refer to the *Sales and Service Negotiation and Quality (SSNQ) User Requirements*.

4.7 System Data

DATA supporting TIPS processing will come from the following sources:

- TIPS Knowledge Base files and Rule files collected as part of the knowledge engineering effort for the TIPS project
 - This information feeds the “reasoning” portion of TIPS processing.
 - The knowledge base files specify Customer Profile items, and potentially, additional product information needed to support the requirements specified in the Functional Requirements.
 - The rule files specify the mapping between Profile Items and Products and Services, and are used to determine specific recommendations to each customer. The information in these files, especially the rules, is dynamic in nature, and may be modified over the lifetime of the system to incorporate new products or to fine-tune the generation of recommendations in light of new market research.
- TIPS Local Customer Database
 - As a temporary measure, TIPS will maintain tables in a Local Database in order to store Customer Profile Information, and any other information relevant to a particular sales session (such as what products were recommended by TIPS, and what sales resulted from the session). This information will be retrieved during repeat calls by the Customer so that the CCEs should not have to request that information again. Until the final set of Profile Items is determined, the definition of the database table may need to change.
 - Ultimately, it is envisioned that the Customer Profile Information collected through TIPS will be stored in the Customer CSDB. The information in these database tables will be available for analysis by Marketing and Product Developers, as specified in the Functional Requirements for this system.
- SSNQ
 - In many cases much of the information about the customer needed by TIPS will have already been retrieved by SSNQ for its own use.
 - In addition to supplying information collected from the customer during the current session, SSNQ provides the access to data resident in Corporate Subject Data Bases (CSDBs) or other existing

- systems.
- All interactions with these systems are mediated by SSNQ. TIPS will not directly access any of these systems.

4.8 Legal Requirements

The legal requirements will be similar to SSNQ. For details, refer to the *Sales and Service Negotiation and Quality (SSNQ) User Requirements*.

4.9 Design Constraints

The overall design of TIPS will be constrained by SSNQ. The user interface component will conform to the view/model/controller model of SSNQ. Because of the limited screen space available and the requirement that TIPS conform to the underlying architecture of SSNQ, there will be a trade-off between usability and ease of implementation.

The necessary information about existing services of customers, product compatibility, availability, definition, benefits and rate quotes is available through the data services provided by SSNQ. For details, refer to the *Sales and Service Negotiation and Quality (SSNQ) Functional Specification*.

4.10 Deployment Requirements

The system will be deployed with SSNQ system. Deployment schedule will follow the schedule of SSNQ. A prototype will be used in the living laboratory trial at Denver in December, 1993. The prototype will be functional if data services and user interface development support from SSNQ are available for TIPS on a timely and consistent manner. Given these conditions are fulfilled, Human Factors Engineers will perform extensive usability studies to ensure that the system meets usability requirements. Studies will include but not limit to the following:

- The quality of the sales tips generated will be studied. Trial scenarios will be developed and compared to the recommendations of the subject matter experts (marketing, CCEs and product management). Metrics can be developed to measure the accuracy, reliability and validity of the knowledge base and the rules used.
- Usability of the customer panel, the account profile, and the needs panel or panels will be studied. Tests will be conducted on the time and effort of CCEs to gather and complete customer profile information.
- Overall usability of other modules like the advertisement, special and office blitz information.

4.11 Documentation and Training Requirement

Documentation and training requirement will be similar to SSNQ. For details, refer to the *Sales and Service Negotiation and Quality (SSNQ) Documentation and Training Plan*.

4.12 Change Administration

Before formally reviewing and agreeing upon this document in June of 1993, the responsible document approval people will review it. After this review, any additional requirements, excluding clarification of information or correction of information in the existing requirements, will be considered for future releases of the system.

The established change management process will be used to request documentation changes. The Project Manager, representative from Carnegie Group and the IT Project Lead must approve all requests for document changes.

5. Issues

(TBD List and describe any points or matters of discussion, debate, or dispute that impact the project that have not been resolved.)

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6. References

The following documents are referred to in this user requirements document:

Sales and Service Negotiation and Quality (SSNQ) User Requirements
U S WEST Communications Information Technologies
IT-CCA-SSNQ-00D0-10-01-00
July 22, 1992

Sales and Service Negotiation and Quality (SSNQ) Functional Specification
U S WEST Communications Information Technologies
(DRAFT SSNQ Internal Document)
IT-CCA-SSNQ-00D0-11-01-00
April 1, 1993

Sales and Service Negotiation and Quality (SSNQ) Documentation and Training Plan
U S WEST Communications Information Technologies
IT-CCA-SSNQ-00D0-50-01-00
April 1, 1993

Appendices

Include an appendix for any detailed information that you reference but do not place in the main body of this document.

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